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A New Approach of Indonesian University Webometrics Ranking Using Entropy and PROMETHEE II

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Abstract

The webometric ranking system is currently one of the most widely used methods to rank website quality including for university websites. It uses pre-determined weights for the criteria assessed in these method; therefore, raised a question whether the pre-determined weigh is objective. Entropy method provides a more objective weighing method criterion. This study aims to compare the use of the entropy method followed by PROMETHEE II methods to evaluate 27 Indonesian university websites. The entropy method assesses four indicators: visibility, presence, openness and excellence of the website performance. The PROMETHEE II method was conducted to provide a complete ranking of university websites from the best to the worst, based on a pair-wise comparison of alternatives along each recognized criterion. The entropy method resulted in 0,3; 0,17;0,23; and 0,27 for visibility, presence, openness and excellence weights, respectively. The website ranking using PROMETHEE II method was not significantly different from the ranking generated from the webometric ranking system. To conclude, the use of entropy as a more objective method followed by PROMETHEE II is a reasonable alternative of the webometric system for ranking university websites, at least in the Indonesian context.

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1. Introduction

Indonesian Higher education institutions have extensive efforts with the goal of improving quality. One of the efforts is having accreditation or certification on the basis of external quality assessment to verify compliance on specific standards and criteria. QS “World University Ranking”, Academic Ranking of World Universities (ARWU), the CWTS Leiden Ranking and the Webometrics Ranking are four university ranking systems that are accepted as a global approach. Webometrics is the result of a quantitative study where the webs are analyzed using essentially four types of indicators, which are : visibility, presence, openness and excellence [1]. Current methods of evaluating universities have various biases and drawbacks. To obtain a correct classification requires very expensive processes in resources and time. First, however, it is necessary to cite some definitions of what is understood in the international literature by webometrics and to examine the webometrics indicators most used by the researchers who have been dedicating themselves to working with this new field of Information Science. In this sense, it should be noted that webometrics arose with the need to measure the information that began to be made available in the network, with the advent of new technologies. It can therefore be defined as a quantitative method of analyzing information available on the World Wide Web, comparable to the already known fields of bibliometrics, scientometrics and informetrics, used to measure information in print and other traditional media. Among the first authors to refer to this new metric are Almind and Ingwersen [2]. They define webometrics as "the application of computational methods to the World Wide Web". For them, the new method aims to investigate communication models, the identification of areas of research, historical studies on the development of a discipline or domain and the evaluation of research by countries, institutions or individuals. Other researchers who have also contributed to the establishment of the first definitions in this field are Abraham and Foresta [3]. For them, webometrics is "an original technique that is born with the purpose of elaborating cognitive maps and mathematical models of the WWW to understand its functioning as a model of social, cultural and political organization of cyberspace." Lennart Björneborn [4] defined web-design as the "study of the quantitative aspects of the construction and use of information resources, structures and technologies on the Web, using bibliometric and computational approaches". The objective of this webometrics ranking is, therefore, to measure the activity and visibility of universities as indicators of their impact and prestige. The position in this method of classification summarizes the overall performance of the university, provides information for students and teachers and reflects the academic commitment to the dissemination of scientific knowledge. The ranking is elaborated from the data published in the open web, not in the intranets, that are indexed by the search engines. In this way, presence and impact web indicators allow an evaluation of university activities, the services offered, the quality of teaching and research, and the importance of scientific, technological, economic and cultural results are transmitted to society.

This paper presents a webometrics analysis of the main Indonesian university sites based on the method of Entropy and Promethee. The main purpose is to find out which sites stand out because they have the highest number of page, links received, and publications. To perform the evaluation of webometrics ranking, one possibility is to combine one method to obtain the criteria weight values. In this sense, an "objective" Entropi method, can be combined that generates point weights with PROMETHEE Method in particular, admit the existence of incomparable alternatives. This leads the decision maker to investigate under what criteria the alternatives show good behavior and under which their performance is deficient, then proceed to make the best choice according to their own scheme of preferences.

2. Method

The method for obtaining the weights of criteria of a new webometrics ranking is the entropy method. After determining the weights in interval form by using the proposed method, then must be followed by ranking. This research used an integrated PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluations) methodology to benchmark a set of websites. PROMETHEE is an outranking method for a finite set of alternative actions to be ranked and selected among criteria, which are often conflicting. It also allows the use of quantitative as well as qualitative criteria when evaluating alternatives. The variables that are measured;

- Impact / visibility: Number of external networks that have backlinks to the institutions' web pages
- Presence: Number of pages of the main web domain of the institution. It includes all subdomains that share the web domain and all types of files including PDF.
- Openness: Number of citations from the main authors by Google Scholar citations.
- Excellence: Number of academic articles published in international journals of high impact among the 10% most cited in their respective scientific disciplines. The data provider (Grup Scimago) data for five years (2010-2014).

The visibility was assessed using the majestic SEO website. The presence of the sites was determined using the advanced search of the Google search engine. The openness and excellence of the websites were calculated using the Google Scholar and Scopus Database Journal, which allowed an estimation of citation and number of journal based on institution. This research examined 27 university website in Indonesia. This data of webometrics parameters from university websites were taken more than 5 trials on various occasions on the different period of time. After the data was collected, it was then organised and analysed and compared with the ranking of Webometrics issued in July 2016. The data was analysed by using a non-parametric statistical test, which is Friedman test in order to analyse whether there are differences among the ranking composition methods.

2.1. Entropi

The entropy method was developed as an objective method of allocation weights depending on the decision matrix without affecting the preference of the decision maker [5], the relative importance of criterion j in a decision situation, w_j measure its weight is directly related to the amount of information provided by the intrinsically set of alternatives with respect to that criterion [6]. How much have greater diversity in the evaluations of the alternatives greater importance should be the criterion. Far this diversity is conceptually based on solid and accepted concept of entropy in an information channel posed by Claude Shannon [7]. The procedure is as follows:

a. The evaluations ij (i = 1, m) (j = 1, n) are taken as normalized as a fraction of the sum i ij Σ to the original assessments of each criterion j.

$$a_{ij} = \frac{k_{ij}}{\sum_{i=1}^{m} \sum_{j=1}^{n} k_{ij}} \quad \text{for } m > 1 \text{ and } i=1, 2, \ldots, m; \text{ and } j=1, 2, \ldots, n. \tag{1}$$

b. Entropy (Ej) is calculated.

$$E_j = [\frac{-1}{\ln(m)}] \sum_{i=1}^{m} [a_{ij} \ln(a_{ij})] \tag{2}$$

where m = number of alternatives in the matrix standardized assessments and ij = Criteria or standardized attributes.

c. Diversity criterion (Dj) is calculated.

$$D_j = 1 - E_j \tag{3}$$

d. The normalized weight of each criterion (Wj) is calculated.

$$w_j = \frac{D_j}{\sum D_j} \tag{4}$$

2.2. PROMETHEE II

In this paper, the PROMETHEE II method is employed to obtain the full ranking of the alternative e-government websites. The procedural steps as involved in PROMETHEE II method are enlisted as below [8]:

Step 1: Normalize the decision matrix using the following equation:

$$R_{ij} = \frac{\left[X_{ij} - \min(X_{ij})\right]}{\left[\max(X_{ij}) - \min(X_{ij})\right]} \quad (i = 1,2,..., n;\ j = 1,2,..., m) \tag{5}$$

$$R_{ij} = \frac{\left[\max(X_{ij}) - X_{ij}\right]}{\left[\max(X_{ij}) - \min(X_{ij})\right]} \tag{6}$$

Step 2: Calculate the evaluative differences of ith alternative with respect to other alternatives. This step involves the calculation of differences in criteria values between different alternatives pair-wise.
Step 3: Calculate the preference function, $P_j(i,i^i)$.

There are mainly six types of generalized preference functions as proposed by Brans [9], but these preference functions require the definition of some preferential parameters, such as the preference and indifference thresholds. However, in real time applications, it may be difficult for the decision maker to specify which specific form of preference function is suitable for each criterion and also to determine the parameters involved. To avoid this problem, the following simplified preference function is adopted here:

$P_{ij}(i,i^1) = 0$ if $R_{ij} < R_i 1_j$ (7)

$Pij(i,i^1) = 1$ if $R_{ij} > R_i 1_j$ (8)

Step 4: Calculate the aggregated preference function taking into account the criteria weights
Aggregated preference function

$$\pi i, i^1 = \left[\sum_{j=1}^{m} W_j X P_j(i, i^1)\right] / \sum_{j=1}^{m} W_j \tag{9}$$

where w_j is the relative importance (weight) of jth criterion.

Step 5: Determine the leaving and entering outranking flows as follows:
Leaving (or positive) flow for i^{th} alternative,

$$\varphi(i) = \frac{1}{n-1} \sum_{i=1}^{n} \pi(i,i) \ \ (i \neq i) \tag{10}$$

Entering (or negative) flow for i^{th} alternative

$$\varphi(i) = \frac{1}{n-1} \sum_{i=1}^{n} \pi(i,i) \ \ (i \neq i) \tag{11}$$

where n is the number of alternatives.

Here, each alternative faces (n – 1) number of other alternatives. The leaving flow expresses how much an alternative dominates the other alternatives, while the entering flow denotes how much an alternative is dominated by the other alternatives. Based on these outranking flows, the PROMETHEE I method can provide a partial preorder of the alternatives, whereas, the PROMETHEE II method can give the complete preorder by using a net flow, though it loses much information of preference relations.

Step 6: Calculate the net outranking flow for each alternative.

$$\varphi(i) = \varphi^{+}(i) = \varphi^{-}(i) \tag{12}$$

Step 7: Determine the ranking of all the considered alternatives depending on the values of φ (i). The higher value of φ (i), the better is the alternative. Thus, the best alternative is the one having the highest φ (i) value. The PROMETHEE method is an interactive multi-criteria decision-making approach designed to handle quantitative as well as qualitative criteria with discrete alternatives. All qualitative criteria are expressed subjectively in linguistic terms. In this method, pair-wise comparison of the alternatives is performed to compute a preference function for each criterion. Based on this preference function, a preference index for alternative i over i' is determined. This preference index is the measure to support the hypothesis that alternative i is preferred to i'. The PROMETHEE method has significant advantages over the other MCDM approaches, e.g. multi-attribute utility theory (MAUT) and AHP. The PROMETHEE method can classify the alternatives, which are difficult to be compared because of a trade-off relation of evaluation standards as non-comparable alternatives [10, 11]. It is quite different from AHP in that there is no need to perform a pair-wise comparison again when comparative alternatives are added or deleted [12, 13].

3. Result and Discussion

The four numbers of criteria that should typically be considered in selecting the best university website are presence, visibility, openness, and excellence. Table 1 gives number of pages of the main web domain of the institution, it includes all sub domains that share the web domain and all types of files including PDF (presence), number of external networks that have backlinks to the institutions' web pages (visibility), number of citations from the main authors by Google Scholar citations (openness), and number of academic articles published in international journals of high impact among the 10% most cited in their respective scientific disciplines which data is provided by Grup Scimago (2010-2014) (excellence). It shows that among all universities, Universitas Gadjah Mada (UGM) had the highest universities number of webpages (presence) while Universitas Terbuka (UT) had the lowest number of webpages. The calculation of visibility requires number of backlinks to a website (given in Table 1).

Table 1. Webometrics Data Based on Criterion

Num.	University	*Presence*	*Visibility*	*Openness*	*Excellence*
1	UI	2.560.000	182.006	10.166	2.353
2	ITB	1.227.143	64.899	18.210	1.303
3	UGM	2.583.571	81.068	28.646	1.322
4	UNDIP	1.385.714	32.556	10.145	475
5	UNRI	332.571	11.187	1.531	204
6	UB	1.642.857	7.530	7.302	611
7	IPB	2.782.857	41.688	5.997	247
8	UNPAD	453.071	129.457	5.849	360
9	UNS	988.500	4.772	1.734	294
10	Unsyiah	751.643	5.358	2.397	332
11	UK Petra	573.929	4.615	699	120
12	UNHAS	912.214	20.894	1.752	392
13	UNAIR	985.357	12.977	3.032	333
14	UNUD	377.357	9.097	2.614	320
15	UNNES	713.571	2.327	897	43
16	UMY	278.286	336.320	2.337	19
17	UNY	720.357	7.440	4.721	61
18	UAD	196.571	2.168	1.516	58
19	USU	1.658.571	15.429	730	99
20	ITS	944.929	40.727	6.009	299
21	UNAND	824.071	19.839	1.491	281
22	UII	627.857	52.431	2.754	76
23	TRISAKTI	162.429	30.948	1.365	66
24	Unika Atma Jaya	550.429	18.139	1.181	39
25	UNILA	518.929	3.956	992	157
26	UT	152.357	27.703	221	3
27	TEL-U	690.000	3.446	1.598	147

Universitas Muhammadiyah Yogyakarta (UMY) had the highest universities visibility while Universitas Ahmad Dahlan (UAD) had the lowest backlinks. Ranked first in the number of page, include subdomain especially from .pdf and .doc file type is Universitas Gadjah Mada (UGM), followed by Institut Teknologi Bandung (ITB) in the second place. Also, there is a tendency that the total number page has an impact to the number of academic files in website.

The final rank of each criterion by using the entropy weighted method can be seen in Table 2. The obtained values of criterion presence, visibility, openness, and excellence are 0,17; 0,33; 0,23; and 0,27 respectively. Based on the calculation the weights of visibility and excellence criteria are just better than the weight of presence and openness. A small program (for example Excel) can be used to determine the weight of each criteria for a complex calculation. The results of entropy method are given in Table 2.

Table 2. Result of Entropy Weight

Variable	*Presence* (*d1*)	*Visibility* (*d2*)	*Openness* (*d3*)	*Excellence* (*d4*)
Weight	0,17	0,33	0,23	0,27

Table 3. Data Normalization

No.	University	*Presence*	*Visibility*	*Openness*	*Excellence*
1	UI	0,9153	0,5382	0,3499	1,0000
2	ITB	0,4086	0,1877	0,6329	0,5532
3	UGM	0,9242	0,2361	1,0000	0,5613
4	UNDIP	0,4689	0,0909	0,3491	0,2009
5	UNRI	0,0685	0,0270	0,0461	0,0855
6	UB	0,5666	0,0160	0,2491	0,2587
7	IPB	1,0000	0,1183	0,2032	0,1036
8	UNPAD	0,1143	0,3809	0,1980	0,1517
9	UNS	0,3179	0,0078	0,0532	0,1236
10	UNSYIAH	0,2278	0,0095	0,0766	0,1398
11	UK Petra	0,1603	0,0073	0,0168	0,0498
12	UNHAS	0,2889	0,0560	0,0539	0,1655
13	UNAIR	0,3167	0,0323	0,0989	0,1404
14	UNUD	0,0855	0,0207	0,0842	0,1349
15	UNNES	0,2133	0,0005	0,0238	0,0170
16	UMY	0,0479	1,0000	0,0744	0,0068
17	UNY	0,2159	0,0158	0,1583	0,0245
18	UAD	0,0168	0,0000	0,0456	0,0232
19	USU	0,5726	0,0397	0,0179	0,0409
20	ITS	0,3013	0,1154	0,2036	0,1257
21	UNAND	0,2554	0,0529	0,0447	0,1183
22	UII	0,1808	0,1504	0,0891	0,0309
23	TRISAKTI	0,0038	0,0861	0,0402	0,0266
24	Unika Atma Jaya	0,1513	0,0478	0,0338	0,0153
25	UNILA	0,1394	0,0054	0,0271	0,0655
26	UT	0,0000	0,0764	0,0000	0,0000
27	TEL-U	0,2044	0,0038	0,0484	0,0613

3.1. Ranking of University Based on Promethee

The decision matrix from Table 1 was used for conducting the Promethee II analysis. The performance values (Table 1) are normalized in the range of [0, 1] by using Eqs. 1 to make the matrix compatible.

Since the criteria in the webometrics system are based on the webometrics published by the Cybermetrics Lab, a research group of the Spanish National Research Council (CSIC) located in Madrid, and considering that all of them have the different degree of importance with regard to their opinions, this research has adopted weights from the calculation from the entropy method. After normalization, the preference function is calculated by using Eq. 7. Weighted preference function is calculated as per Eqs. 7 and results are given in Table 3. From the data presented in Table 3, the PROMETHEE II methodology is applied. Finally, the positive and negative outranking flows, values for net outranking flow for each alternative were obtained according to Eqs. 10-11 and provided in Table 4. The alternative with higher net outranking flow is chosen as the best alternative.

Table 4 presents the results of the application of the PROMETHEE II methodology in the 27 items evaluated by the collaborators. Thus, the value of the adopted weight for each criterion was, being k the number of variable that form in this case, the criteria.

Table 4. Ranking Results Based on Entropy and Promethee

No	Uni	Leaving Flow	Entering Flow	Net OutRangking Flow	Ranking of Webometrics based on Entropy and Promethee	Ranking Based on Webometrics releasedJuly 2016
1	UI	0,9565	0,0435	0,9131	1	2
2	ITB	0,8804	0,1196	0,7608	3	3
3	UGM	0,9450	0,0550	0,8900	2	1
4	UNDIP	0,7977	0,2023	0,5954	4	7
5	UNRI	0,2777	0,6427	-0,3650	20	254
6	UB	0,6788	0,3212	0,3577	8	5
7	IPB	0,7358	0,2642	0,4715	5	4
8	UNPAD	0,7196	0,2804	0,4392	6	6
9	UNS	0,4408	0,5592	-0,1185	16	329
10	UNSYIAH	0,4785	0,5215	-0,0431	13	10
11	UK Petra	0,2119	0,7792	-0,5673	24	15
12	UNHAS	0,6192	0,3808	0,2385	9	11
13	UNAIR	0,6050	0,3950	0,2100	10	8
14	UNUD	0,4562	0,5438	-0,0877	15	9
15	UNNES	0,1488	0,8423	-0,6935	26	22
16	UMY	0,4750	0,5250	-0,0500	14	24
17	UNY	0,3850	0,6150	-0,2300	18	25
18	UAD	0,1254	0,8658	-0,7404	27	17
19	USU	0,4035	0,5877	-0,1842	17	36
20	ITS	0,7042	0,2958	0,4085	7	12
21	UNAND	0,4831	0,5081	-0,0250	12	29
22	UII	0,5462	0,4538	0,0923	11	20
23	TRISAKTI	0,3377	0,6535	-0,3158	19	32
24	Unika Atma Jaya	0,2823	0,7088	-0,4265	22	46
25	UNILA	0,2335	0,7577	-0,5242	23	13
26	UT	0,2031	0,7881	-0,5850	25	21
27	TEL-U	0,2896	0,7104	-0,4208	21	23

According to this methodology, Universitas Indonesia, Universitas Gadjah Mada and Institut Teknologi Bandung are selected as the best website in webometrics. Based on the Wilcoxon Test calculation, the value of Z obtained at -0.18 with p value (Asymp.Sig. (2-tailed) of 0.861 where the value is greater than the critical limits of research that is 0.05 so that there is no significant difference between the Promethee Ranks group and the Webometrics Ranks group. As a future direction, other decision -making methods such as fuzzy Promethee can be used in this area.

4. Conclusion

This research proposed methodology for determining and evaluating the best university website based on four criteria of website based on PROMETHEE II and Entropy. The results on the webometrics ranking based on sample data from 27 university websites and calculate presence, visibility, openness and excellence shows that the weight of criteria of webometrics using entropy method are different with the weight of webometrics ranking system. This new combination model has been implemented to generate the weights for the criteria which are more objective and guaranteed more fairly preference of criteria. The result of this study confirmed that the ranking website of Indonesian university websites are almost similar with the ranking results produced by webometrics published by the Cybermetrics Lab. The use of entropy as a more objective method followed by PROMETHEE II is a reasonable alternative of the webometric system for ranking university websites, at least in the Indonesian context.

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